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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026

25MCA104 – MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Use set builder notation and logical equivalence to establish De Morgan law of $(A \cap B)' = A' \cup B'$.	10	L2	1	1
b)	If A and B are the matrices, find AB if it is defined.				
	$A = \begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix}$	05	L1	1	1
c)	Discuss the basic operation on sets with Venn diagrams.	05	L2	1	1
2. a)	Find eigen value and eigen vector of the following 2*2 matrix.				
	$A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$	10	L2	1	1
b)	Find the join and meet of the zero-one matrices.				
	$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	05	L1	1	1
c)	Define inclusion exclusion principle with an example.	05	L2	1	1
Unit – II					
3. a)	Discuss the applications of propositional logic.	05	L1	2	1
b)	Translate the following statement into English, where the domain for both variables consists of all real numbers. $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$.	05	L2	2	1
c)	Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.	10	L2	2	1
4. a)	Translate the statement "The sum of two positive integers is always positive" into a logical expression.	10	L2	2	1
b)	Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.	10	L2	2	1
Unit – III					
5. a)	Consider the following relations on $\{1,2,3,4\}$: $R1 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$, $R2 = \{(1,1),(1,2),(2,1)\}$, $R3 = \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)\}$, $R4 = \{(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)\}$, $R5 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)\}$, $R6 = \{(3,4)\}$. Which of these relations are reflexive. Why?	05	L2	3	1
b)	Let A be the set $\{1,2,3,4\}$. Which ordered pairs are in the relation $R = \{(a,b) \mid a \text{ divides } b\}$? Display the result in graphical and tabular form.	05	L2	3	1



c) What results when the following projections are applied to the 4-tuples (2,3,0,4), (Jane Doe, 234111001, Geography, 3.14), and (a1,a2,a3,a4)?

- i) P1,3
- ii) P2,4

10 L1 3 1

6. a) Suppose that the relation R on a set is represented by the matrix below, Whether the relation R is reflexive, symmetric, and/or anti symmetric?

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

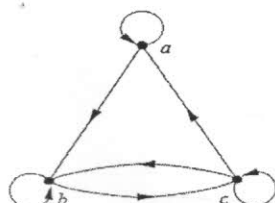
05 L1 3 1

b) Find the zero-one matrix of the transitive closure of the relation R where

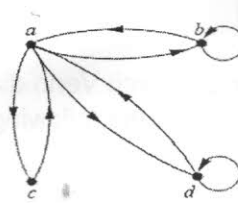
$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

05 L1 3 1

c) Determine whether the relations for the directed graphs shown in Figure a & figure b are reflexive, symmetric, antisymmetric, and/or transitive.



(a) Directed graph of R



(b) Directed graph of S

10 L2 3 1

Unit – IV

7. a) What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled?

10 L2 4 1

b) If X is a random variable on a sample space S, then $V(X) = E(X^2) - E(X)^2$.

10 L2 4 1

8. a) Prove Bayes theorem.

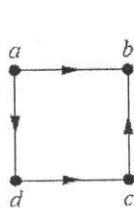
10 L2 4 1

b) A fair coin is flipped three times. Let S be the sample space of the eight possible outcomes, and let X be the random variable that assigns to an outcome the number of heads in this outcome. What is the expected value of X?

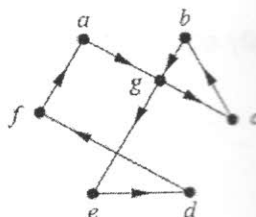
10 L3 4 1

Unit – V

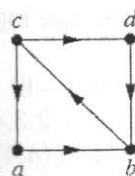
9. a) Which of the directed graphs shown below have an Euler circuit? Which have an Euler path?



H₁



H₂

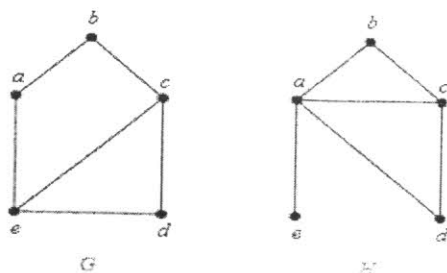


H₃

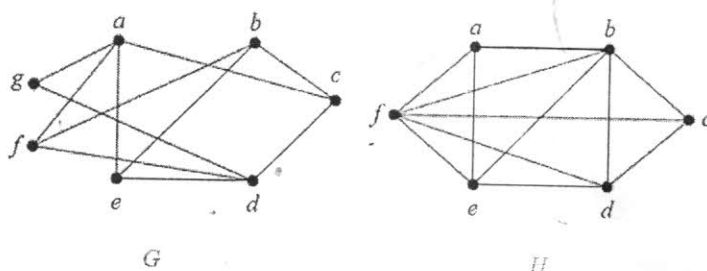
10 L2 5 1

b) Justify whether the graphs displayed below are isomorphic.

10 L3 5 1



10. a) Whether the graphs G and H shown below are bipartite? Give reasons.



- b) Explain each with examples
 i) Adjacency matrices
 ii) Incidence matrices

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
